Topic Name: Neural Networks

A neural network attempts to use input to predict output.

Inputs			Output
X1	X2	X3	Y
0	0	1	0
0	1	1	0
1	0	1	1
1	1	1	1

Weights

GPA example:

Student#1

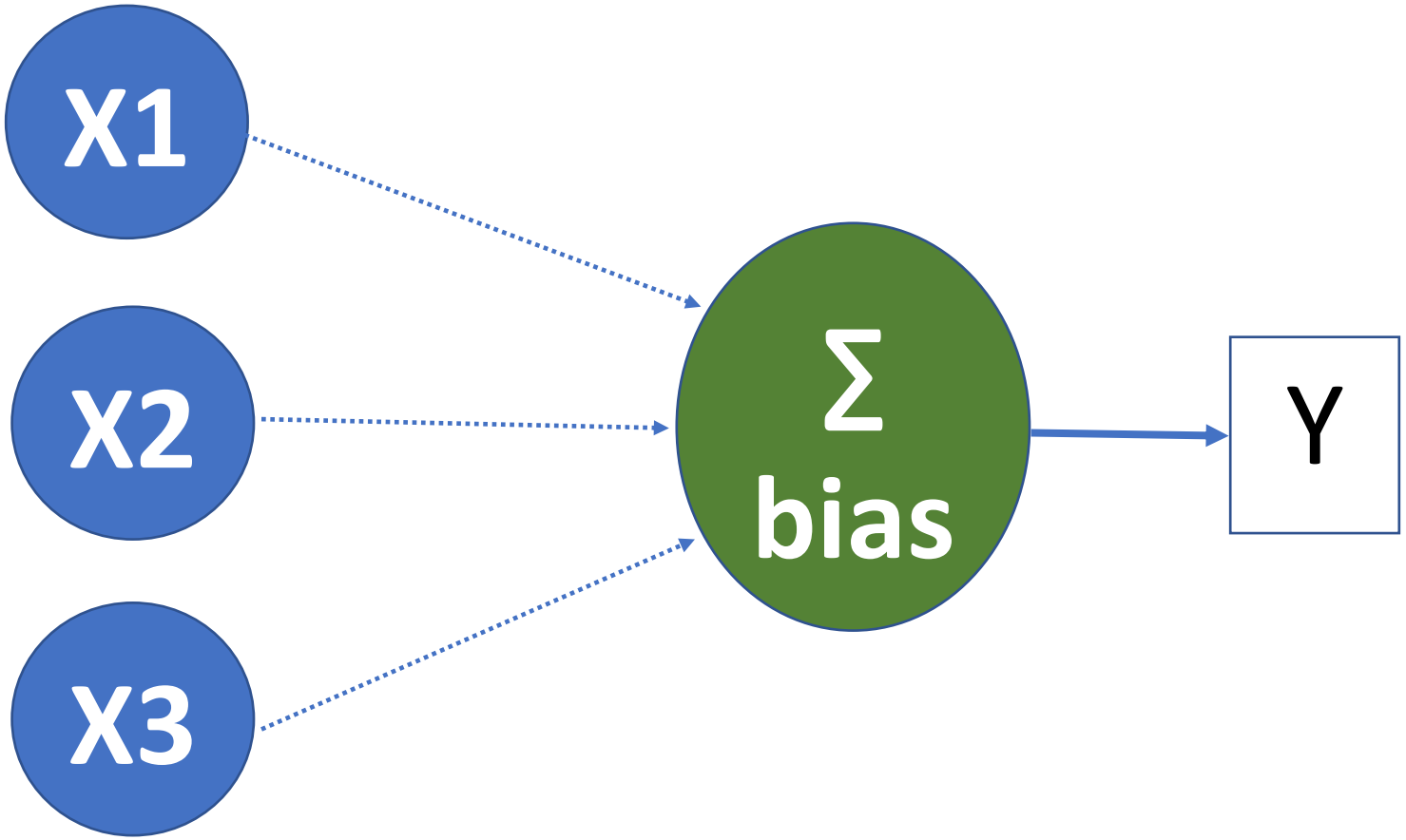
OOP 3 hours A …. Database 3 hours C

GPA = ((4 * 3) + (2 * 3)) / 6 = 3.0

Student#2

OOP 3 hours A …. DB-Lab 1 hour C

GPA = ((4 * 3) + (2 * 1)) / 4 = 3.5



$$Y = X1 * w1 + X2*w2 + X3*w3 + bias$$

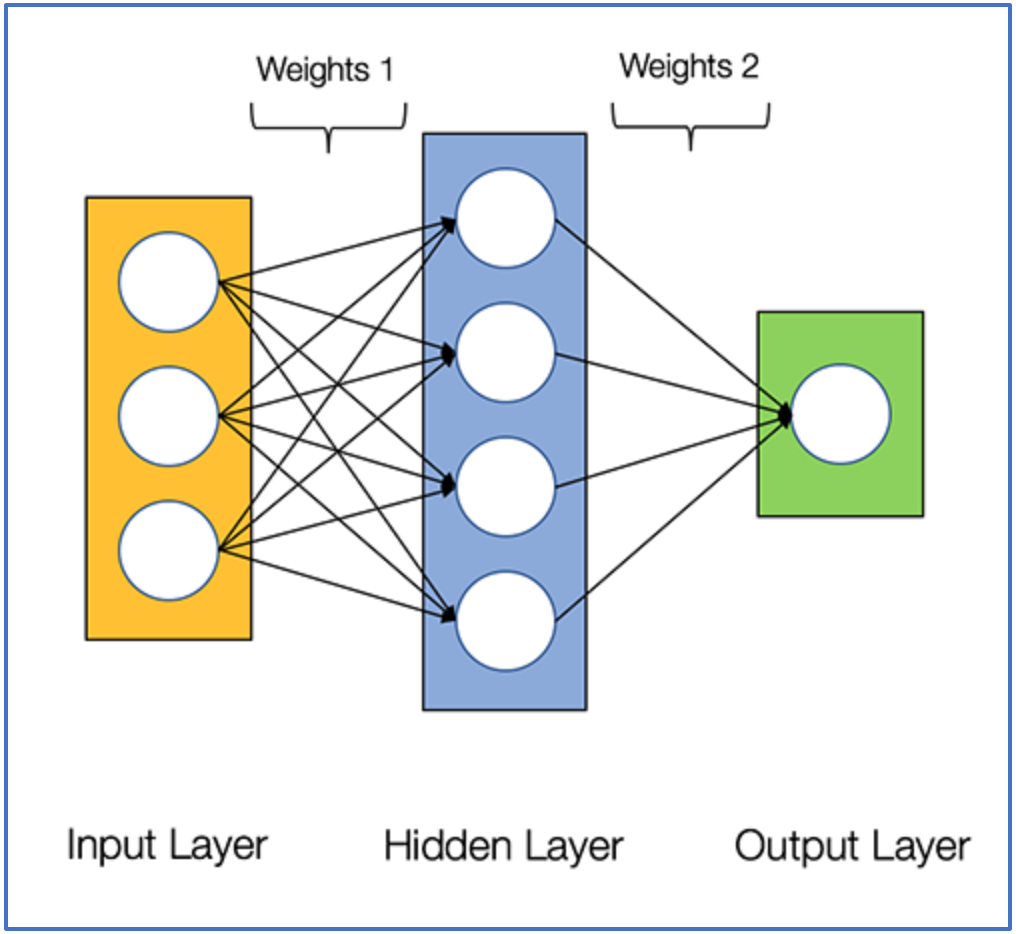
The process of fine-tuning the weights and biases from the input data is known as **training the Neural Network.**

A Slightly Harder Problem

Inputs			Output
X1	X2	X3	Y
0	0	1	0
0	1	1	1
1	0	1	1
1	1	1	0

This is considered a "nonlinear" pattern because there isn't a direct one-to-one relationship between the input and output. Instead, there is a one-to-one relationship between a combination of inputs, namely columns 1 and 2.

In order to first combine pixels into something that can then have a one-to-one relationship with the output, we need to add another layer. Our first layer will combine the inputs, and our second layer will then map them to the output using the output of the first layer as input.


Weights 1
Weights 2
Input Layer
Hidden Layer
Output Layer

Each iteration of the training process consists of the following steps:

- Calculating the predicted output, known as **feedforward**
- Updating the weights and biases, known as **backpropagation**

However, we still need a way to evaluate the "goodness" of our predictions (i.e. how far off are our predictions)? The **Loss Function** allows us to do exactly that.

Loss Function

There are many available loss functions, and the nature of our problem should dictate our choice of loss function. A simple **sum-of-squares error** can be used as our loss function.

$$Sum-of-Squares\ Error = \sum_{i=1}^{n} (y - \hat{y})^2$$

That is, the sum-of-squares error is simply the sum of the difference between each predicted value and the actual value. The difference is squared so that we measure the absolute value of the difference.

Our goal in training is to find the best set of weights and biases that minimizes the loss function.